

Structural vs Empirical Diffusion in Boolean Cryptographic Transformations

Abstract

The question being asked: To what extent do structural properties of Boolean transformations predict their empirical diffusion behavior?

We evaluated boolean transformations with varying input and output widths, comprising families, e.g., identity, linear XOR, nonlinear, random, S-box. Structural measures include dependency density, diffusion degree, confusion degree, component algebraic degree; empirical measures include mean avalanche rate, SAC deviation, per-input/per-output flip probabilities. The analysis compares these metrics across functions and within each function's input-output bit pairs.

Overall, our experiments partially support the hypothesis that greater structural dependency is associated with stronger avalanche behavior. While increased dependency often accompanies higher avalanche propagation, it does not necessarily guarantee balanced avalanche behavior or better SAC performance.

1. Motivation and hypothesis

A structural measure asks whether a change in an input bit *can* affect an output bit. An empirical measure asks how frequently it actually does so over tested inputs. These are related questions, but they are not equivalent.

We investigate:

- **H1 - Coverage:** Transformations with higher dependency density generally exhibit a larger *mean fraction* of output bits changed when one input bit flips.
- **H2 - Balance:** High structural diffusion does **not** by itself guarantee the Strict Avalanche Criterion (SAC), which requires each input-output bit pair to flip with probability approximately **0.5**.
- **H3 - Nonlinearity/structure:** Structural confusion and component algebraic degree may help characterize transformations, but neither alone guarantees good avalanche or SAC performance.

2. Mathematical definitions

Let

$$F: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m, \quad F = (f_0, \dots, f_{m-1}).$$

Here, e_i flips only input bit i , and $\oplus$ is XOR.

2.1 Structural measures

Dependency matrix

$$D_{ij} = \begin{cases} 1, & \exists x: f_j(x) \neq f_j(x \oplus e_i), \\ 0, & \text{otherwise.} \end{cases}$$

Dependency density:

$$\rho_D = \frac{1}{nm} \sum_{i=0}^{n-1} \sum_{j=0}^{m-1} D_{ij}.$$

Structural diffusion degree:

$$d(F) = \min_i \sum_{j=0}^{m-1} D_{ij}.$$

It is the **minimum number of output components structurally dependent on any one input bit**. $d(F)$ is reported in output-bit units; if comparing across differing m , we also report $d(F)/m$.

Confusion degree: Number of input variables appearing in the ANF of every output component.

$C(F)$ = No of elements in $\cap_{i=0}^{n-1} \Omega_i$

Component algebraic degree: Maximum degree of a nonzero ANF monomial in f_j

2.2 Empirical measures

For uniform input x (enumerating the full domain, when feasible), define the avalanche-probability matrix

$$A_{ij} = \Pr_x[f_j(x) \neq f_j(x \oplus e_i)].$$

The **mean avalanche fraction** is

$$\bar{A} = \frac{1}{nm} \sum_{i,j} A_{ij}.$$

SAC target: every $A_{ij} = 0.5$. To quantify deviation, we report

$$\Delta_{\text{SAC,mean}} = \frac{1}{nm} \sum_{i,j} |A_{ij} - 0.5|, \quad \Delta_{\text{SAC,max}} = \max_{i,j} |A_{ij} - 0.5|.$$

We also calculate a $\text{SAC_score} = 1 - 2 \cdot \Delta_{\text{SAC,mean}}$.

Lower is better for both deviations. A mean avalanche fraction near 0.5 **does not prove** SAC: it can hide some pairs near 0 and others near 1.

2.3 The exact link between structural and empirical dependency

With exhaustive inputs,

$$D_{ij} = 1 \Leftrightarrow A_{ij} > 0.$$

This follows because a dependency exists exactly when at least one input produces a change in output bit j .

Perfect SAC requires $A_{ij} = \frac{1}{2}$, and since all A_{ij} are positive, $D_{ij} = 1$ for all i, j . Thus, $D = 1_{n \times m}$.

Consequently, $d(F) = m$ and $\rho_D = 1$.

This is an important **one-way relationship**: a missing structural dependency **forces** zero avalanche probability, but a present structural dependency says nothing about whether the flip probability is 0.01, 0.5, or 1.0. It follows that perfect SAC requires $D_{ij} = 1$ for all pairs, hence $\rho_D = 1$ and $d(F) = m$. The converse does not hold.

3. Experimental setup and reproducibility

Item	Value
Input width(s), n	[3, 4, 8]
Output width(s), m	[3, 4, 8]
Function families	Hand-designed Boolean transformations, cryptographic S-boxes (AES and PRESENT), and synthetic ANF-generated Boolean transformations
Number of transformations evaluated	28(6 hand-designed + 2 cryptographic S-boxes + 20 synthetic)
Input sampling	Exhaustive evaluation over all (2^n) inputs per input-bit flip in the exhaustive implementation
Random seed(s)	20261008
Raw metrics CSV(s)	results/summary.csv, results/dependency_matrices/, results/avalanche_matrices/, results/anf/
Plot directory	results/plots/

4. Results at a glance

Transformation	Family	$n \rightarrow m$	Density ρ_D	Diffusion $d(F)$	Confusion $c(F)$	ANF degree	Mean avalanche	Mean SAC deviation	Max SAC deviation
Identity	Toy	3→3	0.3333	1	0	1	0.3333	0.5000	0.5000
Linear Mix	Toy	3→3	0.7778	2	1	1	0.7778	0.5000	0.5000
Sample Function	Toy	3→3	0.7778	2	1	2	0.6667	0.3889	0.5000
Quadratic Mix	Toy	3→3	1.0000	3	3	2	0.6667	0.1667	0.5000
Uneven Nonlinear	Toy	3→3	0.7778	2	1	2	0.6667	0.3889	0.5000
Cubic Mix	Toy	3→3	1.0000	3	3	3	0.5833	0.2500	0.2500
PRESENT S-box	Cryptographic	4→4	1.0000	4	4	3	0.6250	0.1250	0.5000
AES S-box	Cryptographic	8→8	1.0000	8	8	7	0.5049	0.0264	0.0625
Sparse linear 1	Synthetic	4→4	0.3125	1	0	1	0.3125	0.5000	0.5000
Sparse linear 2	Synthetic	4→4	0.3125	1	0	1	0.3125	0.5000	0.5000
Sparse linear 3	Synthetic	4→4	0.3125	0	0	1	0.3125	0.5000	0.5000

Sparse linear 4	Synthetic	4→4	0.4375	0	0	1	0.4375	0.5000	0.5000
Dense linear 1	Synthetic	4→4	0.7500	2	1	1	0.7500	0.5000	0.5000
Dense linear 2	Synthetic	4→4	0.5625	1	0	1	0.5625	0.5000	0.5000
Dense linear 3	Synthetic	4→4	0.5625	2	0	1	0.5625	0.5000	0.5000
Dense linear 4	Synthetic	4→4	0.4375	1	0	1	0.4375	0.5000	0.5000
Sparse quadratic 1	Synthetic	4→4	0.3750	1	0	1	0.3750	0.5000	0.5000
Sparse quadratic 2	Synthetic	4→4	0.5000	2	0	2	0.3750	0.3750	0.5000
Sparse quadratic 3	Synthetic	4→4	0.5000	1	0	2	0.2813	0.2813	0.5000
Sparse quadratic 4	Synthetic	4→4	0.6250	2	0	2	0.4375	0.3125	0.5000
Medium quadratic 1	Synthetic	4→4	0.9375	3	3	2	0.5313	0.0938	0.5000
Medium quadratic 2	Synthetic	4→4	0.8750	3	2	2	0.5000	0.1250	0.5000
Medium quadratic 3	Synthetic	4→4	0.9375	3	3	2	0.4688	0.0313	0.5000
Medium quadratic 4	Synthetic	4→4	0.8125	3	1	2	0.5625	0.2500	0.5000
Mixed cubic 1	Synthetic	4→4	0.8125	3	1	3	0.6094	0.3281	0.5000
Mixed cubic 2	Synthetic	4→4	0.9375	3	3	3	0.4219	0.1406	0.5000
Mixed cubic 3	Synthetic	4→4	0.7500	2	1	3	0.5938	0.3438	0.5000
Mixed cubic 4	Synthetic	4→4	0.7500	2	1	2	0.5313	0.2813	0.5000

Immediate observations:

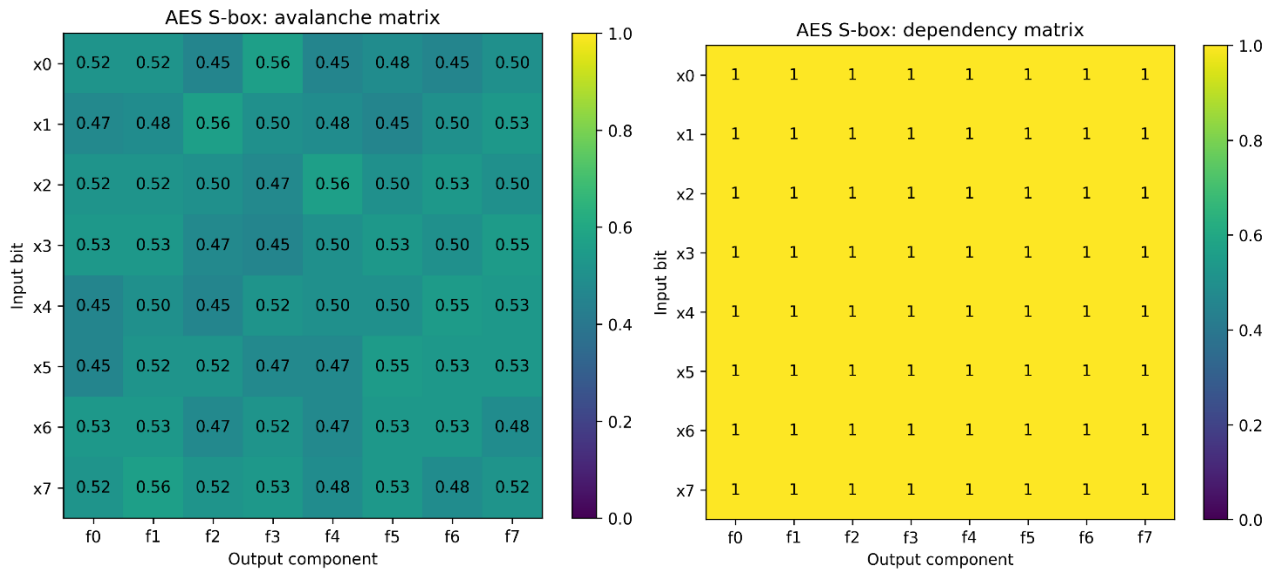
1. **Highest structural coverage:** Quadratic Mix, Cubic Mix, PRESENT S-Box, AES S-box with $\rho_D = 1$, $d/m = 1$; The AES S-box exhibits particularly balanced empirical $\bar{A} = 0.5049$ and SAC deviation 0.0264. By contrast Quadratic Mix has $\bar{A} = 0.6667$ and mean SAC deviation = 0.1667 despite having complete structural coverage.
2. **Best SAC balance:** AES S-box with lowest mean SAC deviation 0.0264 and lowest maximum SAC deviation 0.0625. Among synthetic transformations, Medium quadratic 3 has the lowest mean SAC deviation 0.0313, although its maximum deviation remains 0.5000, indicating that at least one input-output pair is completely imbalanced.
3. **Largest structural/empirical mismatch:** Quadratic Mix has $\rho_D = 1$ but $\bar{A} = 0.6667$, SAC deviation 0.1667 and maximum SAC deviation = 0.5000. Dense linear 1 offers another important example: despite having dependency density 0.7500 and mean avalanche probability 0.7500, its mean SAC deviation is 0.5000.

Interpretation: The results demonstrate that structural diffusion and empirical avalanche balance capture fundamentally different properties of Boolean transformations. Complete structural dependency ensures that each input bit is capable of influencing every output bit, but does not guarantee that these changes occur with probability 0.5. Overall, the experiments indicate that structural metrics are informative measures of potential bit influence, but empirical avalanche measurements are necessary to assess its probabilistic balance.

5. Figure-by-figure analysis

5.1 Dependency heatmap vs avalanche-probability heatmap

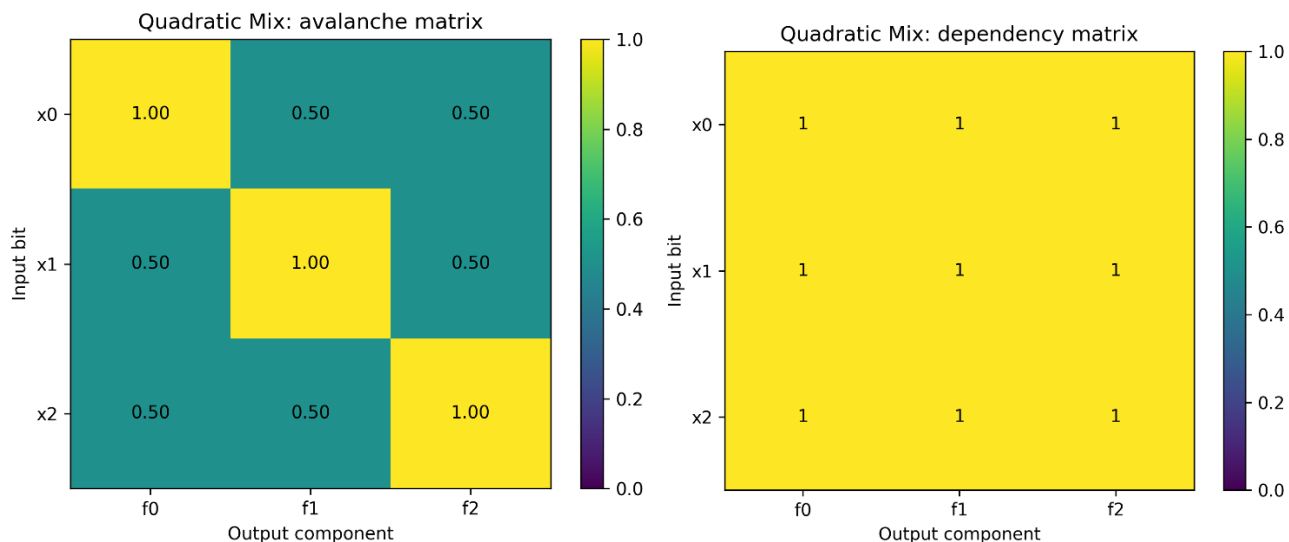
- **AES S-box**



The AES S-box dependency matrix contains 64 active input-output pairs out of 64 possible pairs, giving a dependency density of 1.0000 and diffusion degree of 8. This establishes complete structural dependency, as every input bit is capable of influencing every output component.

Of these 64 active pairs, 64 (100%) have avalanche probabilities in the interval $[0.45, 0.57]$, as reflected in the heatmap. The observed probabilities are concentrated around the ideal SAC value of 0.5, with a mean avalanche probability of 0.5049, mean SAC deviation of 0.0264, and maximum SAC deviation of 0.0625.

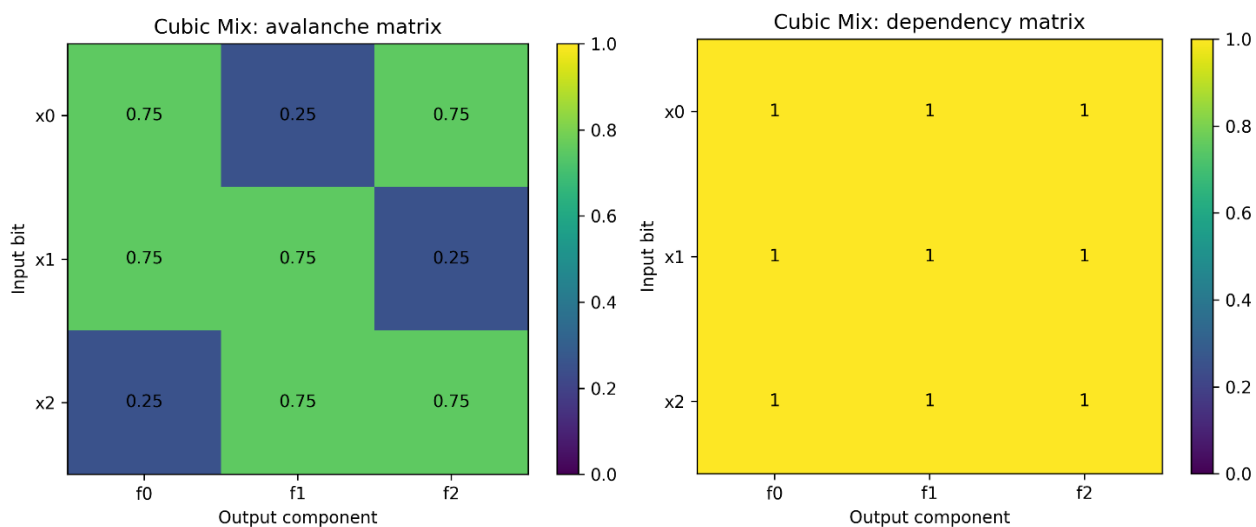
- **Quadratic Mix**



The Quadratic Mix dependency matrix contains 9 active input-output pairs out of 9 possible pairs, giving a dependency density of 1.0000, diffusion degree of 3, and confusion degree of 3. This establishes complete structural dependency, as every input bit is capable of influencing every output component.

Of these 9 active pairs, 6 (66.67%) have avalanche probabilities equal to 0.5, while the remaining 3 (33.33%) have avalanche probabilities equal to 1.0. The transformation exhibits a mean avalanche probability of 0.6667, mean SAC deviation of 0.1667, and maximum SAC deviation of 0.5000.

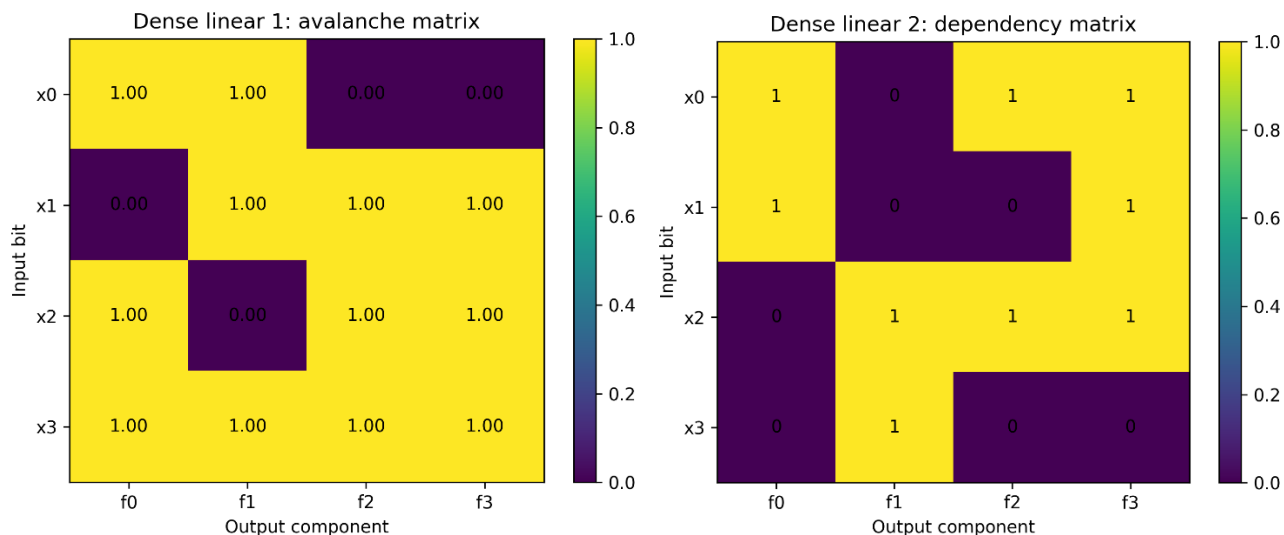
- **Cubic Mix**



The Cubic Mix dependency matrix contains 9 active input-output pairs out of 9 possible pairs, giving a dependency density of 1.0000, diffusion degree of 3, and confusion degree of 3. This establishes complete structural dependency, as every input bit is capable of influencing every output component.

Of these 9 active pairs, 6 (66.67%) have avalanche probabilities equal to 0.75, while the remaining 3 (33.33%) have avalanche probabilities equal to 0.25. None of the input-output pairs satisfy the Strict Avalanche Criterion exactly, since no avalanche probability equals 0.5. The transformation exhibits a mean avalanche probability of 0.5833, mean SAC deviation of 0.2500, and maximum SAC deviation of 0.2500.

- **Dense Linear 1**



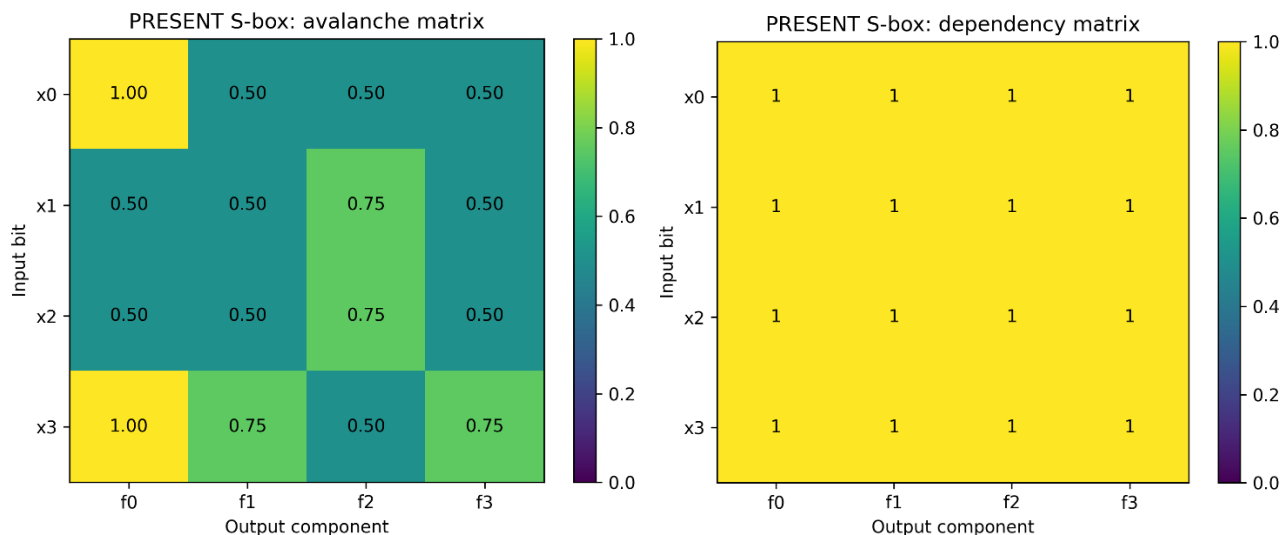
Dense Linear 1 has 12 active input-output pairs out of 16 possible pairs, giving a dependency density of 0.7500, diffusion degree of 2, and confusion degree of 1. This indicates relatively high structural connectivity, although some input-output dependencies are absent.

Of the 16 possible input-output pairs, 12 (75%) have avalanche probabilities equal to 1.0, while the remaining 4 (25%) have avalanche probabilities equal to 0.0. None of the pairs satisfy the Strict Avalanche Criterion, since no avalanche probability equals 0.5. The transformation exhibits a mean avalanche probability of 0.7500, mean SAC deviation of 0.5000, and maximum SAC deviation of 0.5000.

The avalanche heatmap exhibits a binary pattern consisting exclusively of probabilities 0 and 1. This behaviour is characteristic of linear Boolean transformations, where flipping a particular input bit either always changes a given output component or never changes it, independently of the original input vector

These observations demonstrate that high average avalanche probability does not necessarily imply balanced avalanche behaviour. The results therefore highlight an important limitation of relying on mean avalanche probability as an independent measure of diffusion quality.

- **PRESENT S-box**



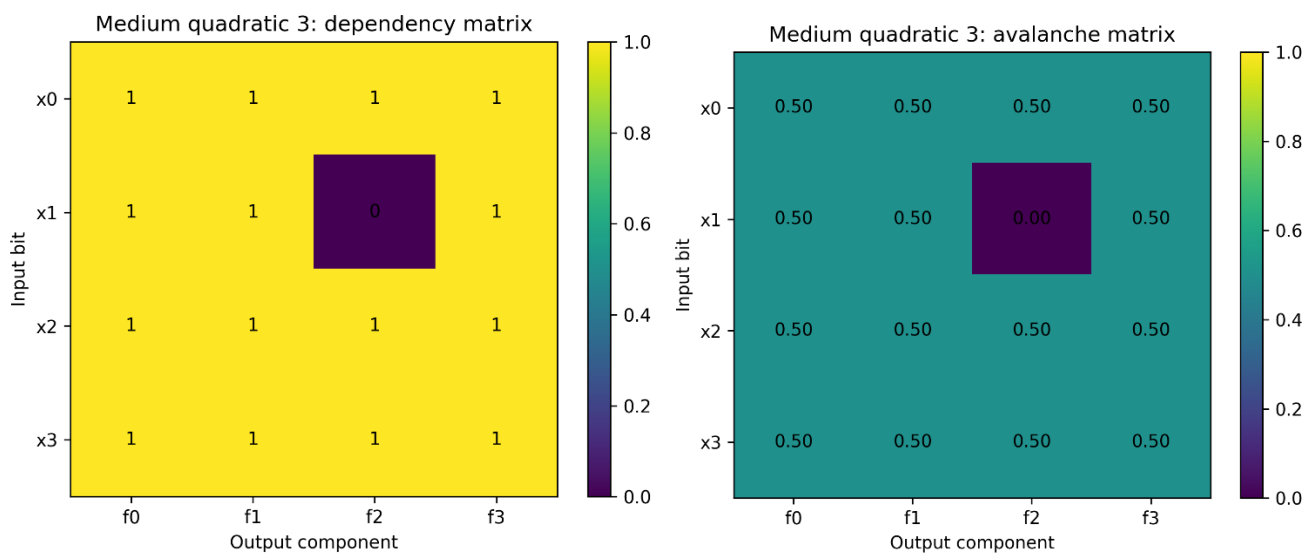
According to the experimental results, Dense Linear 1 has 12 active input-output pairs out of 16 possible pairs, giving a dependency density of 0.7500, diffusion degree of 2, and confusion degree of 1. This indicates relatively high structural connectivity, although some input-output dependencies are absent.

Of the 16 possible input-output pairs, 12 (75%) have avalanche probabilities equal to 1.0, while the remaining 4 (25%) have avalanche probabilities equal to 0.0. None of the pairs satisfy the Strict Avalanche Criterion, since no avalanche probability equals 0.5. The transformation exhibits a mean avalanche probability of 0.7500, mean SAC deviation of 0.5000, and maximum SAC deviation of 0.5000.

These observations demonstrate that the PRESENT S-box achieves maximum structural diffusion and confusion according to the implemented metrics but does not satisfy perfect SAC.

The results therefore demonstrate that even established cryptographic S-boxes with maximum structural dependency can exhibit substantially different empirical avalanche characteristics.

- **Medium Quadratic 3**



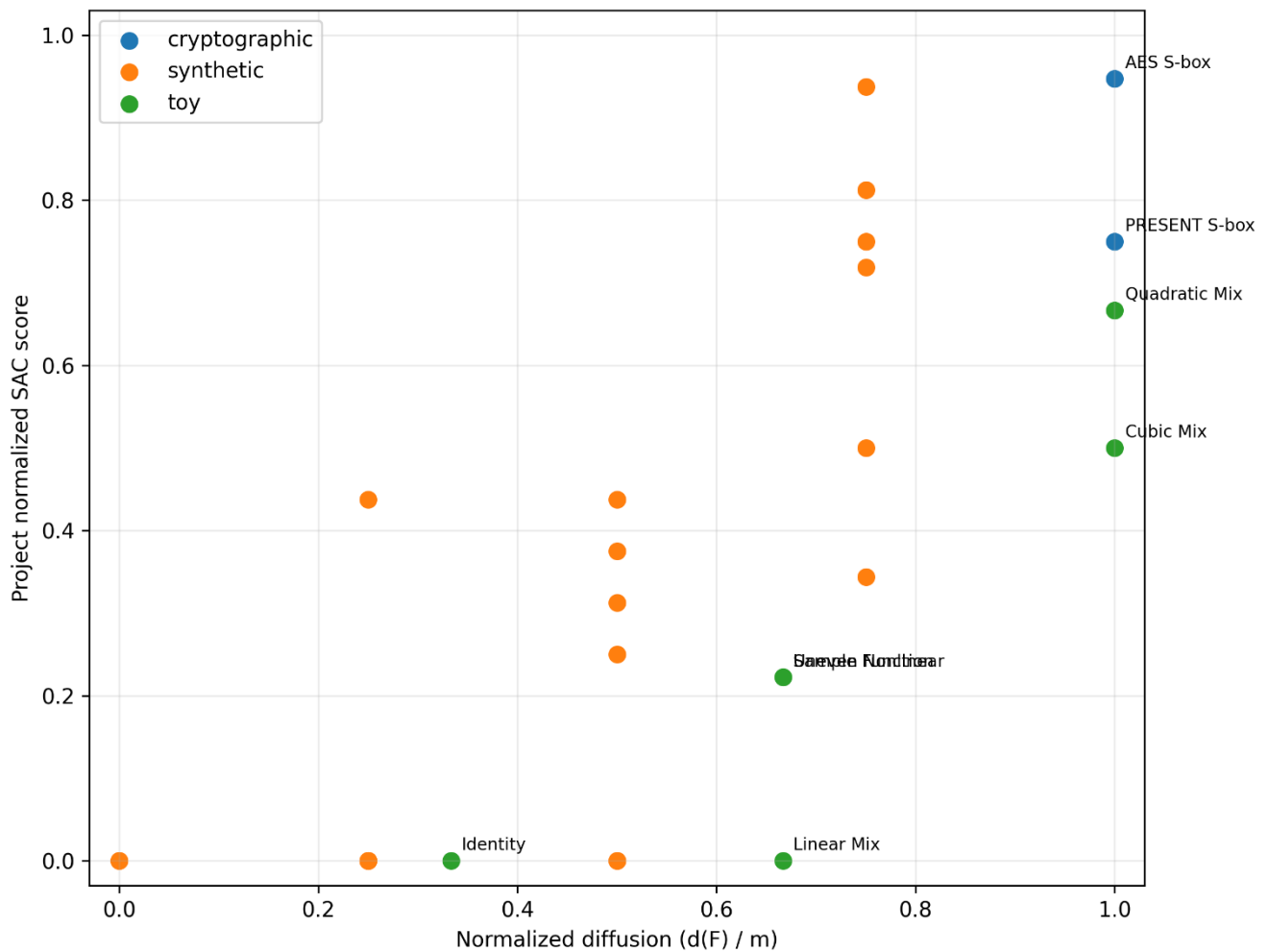
Medium Quadratic 3 has 15 active input-output pairs out of 16 possible pairs, giving a dependency density of 0.9375, diffusion degree of 3, and confusion degree of 3. This indicates extensive structural connectivity, with only one input-output pair lacking dependency.

The transformation exhibits a mean avalanche probability of 0.4688, mean SAC deviation of 0.0313, and maximum SAC deviation of 0.5000. The low mean SAC deviation indicates that the avalanche probabilities are, on average, close to the ideal value of 0.5. However, the maximum deviation reveals that at least one input-output pair exhibits a completely imbalanced response.

Thus, 15 out of 16 pairs (93.75%) satisfy SAC exactly, but the transformation as a whole fails perfect SAC because one output component is structurally independent of a particular input bit.

This comparison demonstrates that a low average SAC deviation can conceal a severe weakness in an individual input-output dependency. Therefore, mean SAC deviation should be interpreted alongside maximum deviation and the full avalanche matrix

5.3 Diffusion vs empirical behavior



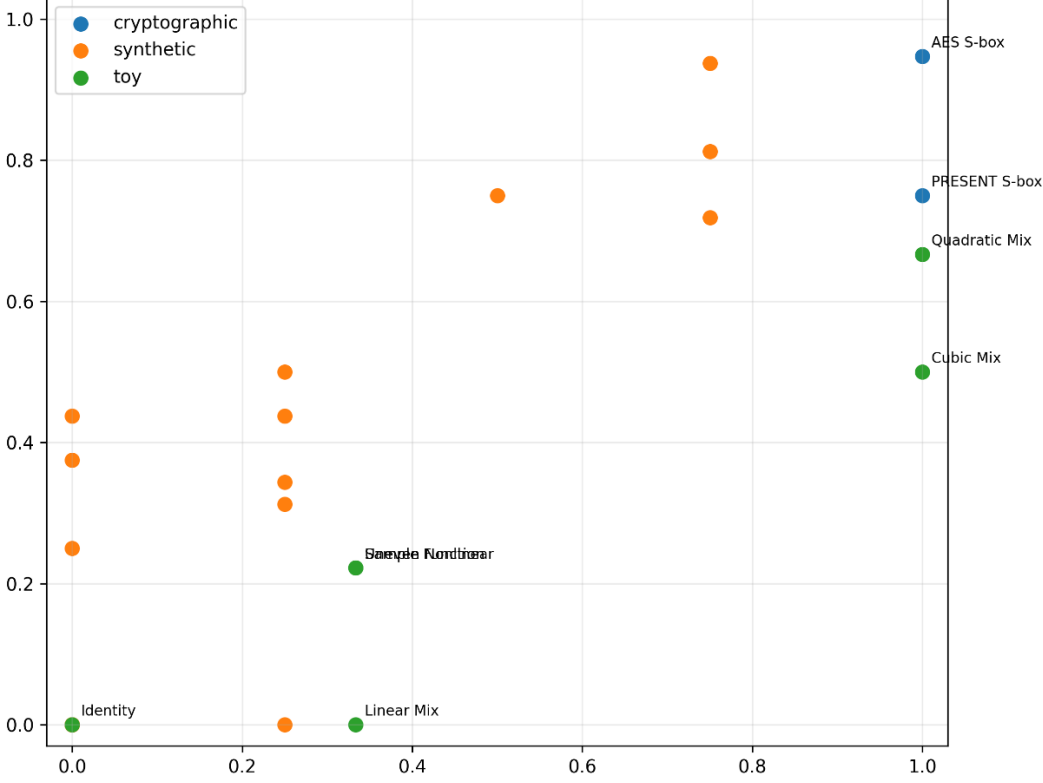
The scatter plot compares normalized diffusion degree ($d(F)/m$) against normalized SAC score across 28 Boolean transformations.

The results suggest that higher structural diffusion is often accompanied by improved SAC performance, particularly among nonlinear transformations. However, the relationship is not strictly monotonic, and transformations with identical diffusion degrees can exhibit substantially different avalanche behaviour.

- Maximum diffusion does not guarantee optimal SAC: AES S-box, PRESENT S-box, Quadratic Mix, and Cubic Mix all achieve normalized diffusion of 1.0000, yet their SAC scores differ considerably: 0.9473, 0.7500, 0.6667, and 0.5000, respectively.
- High diffusion can coexist with poor avalanche balance: Linear Mix achieves normalized diffusion of 0.6667 but has a SAC score of 0.0000, demonstrating that substantial structural connectivity does not guarantee balanced output changes.
- Similar diffusion degrees can produce different empirical results: Medium Quadratic 3 and Mixed Cubic 1 both achieve normalized diffusion of 0.7500, but their SAC scores are 0.9375 and 0.3438, respectively.

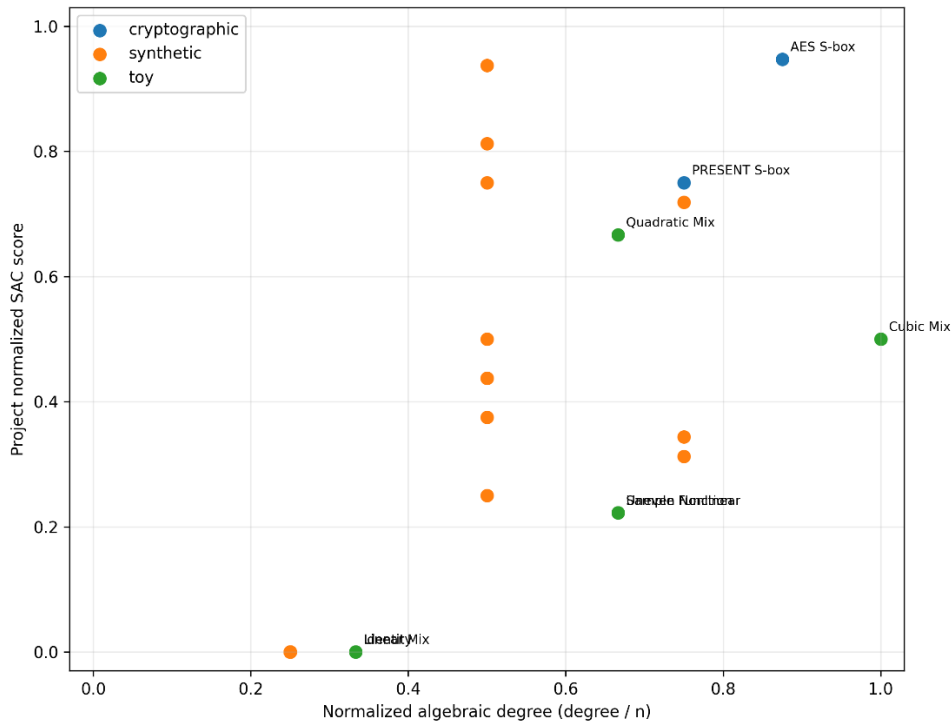
Therefore, structural diffusion and empirical avalanche balance are complementary rather than interchangeable metrics, and both must be considered when evaluating Boolean transformations.

The two scatter plots compare **normalized confusion degree** ($(c(F)/n)$) and **normalized algebraic degree** ($(\deg(F)/n)$) against the normalized SAC score across **28 Boolean transformations**, comprising 6 toy transformations, 20 synthetic transformations, and 2 cryptographic S-boxes.



Our results show that several transformations with higher normalized confusion achieve relatively high SAC scores. However, the relationship is not uniform, and transformations possessing identical confusion values can exhibit substantially different empirical behaviour.

- Maximum confusion does not guarantee optimal SAC behaviour. The AES S-box, PRESENT S-box, Quadratic Mix, and Cubic Mix all attain maximum normalized confusion of 1.0000, yet their normalized SAC scores are 0.9473, 0.7500, 0.6667, and 0.5000, respectively
- Low confusion does not necessarily imply maximum SAC deviation. Sparse Quadratic 3 has normalized confusion 0.0000 but achieves a normalized SAC score of 0.4375.
- Intermediate confusion values exhibit considerable variation. In particular, Mixed Cubic 1 has normalized confusion 0.2500 and SAC score 0.3438, whereas Medium Quadratic 3 reaches normalized confusion 0.7500 and SAC score 0.9375.
- Nonlinear transformations generally exhibit more varied avalanche behavior than linear transformations. Several nonlinear synthetic transformations achieve moderate or high SAC scores, while all evaluated linear transformations have a normalized SAC score of zero



The second scatter plot investigates whether increasing algebraic degree is associated with improved empirical avalanche balance.

Algebraic degree represents the highest-degree monomial appearing in the Algebraic Normal Form (ANF) of a Boolean transformation's output components. A higher degree indicates greater algebraic nonlinearity in this specific sense, but it does not independently establish balanced avalanche behavior or cryptographic security.

- Linear transformations consistently exhibit poor SAC behavior. All evaluated degree-1 transformations achieve normalized SAC scores of 0.0000, regardless of their dependency density.
- Quadratic transformations exhibit substantial variation in SAC performance. Quadratic Mix has algebraic degree 2 and SAC score 0.6667, whereas Medium Quadratic 3, also of degree 2, achieves a substantially higher SAC score of 0.9375
- Higher algebraic degree does not guarantee improved SAC behavior. Cubic Mix has algebraic degree 3 and normalized SAC score 0.5000, while Quadratic Mix has a lower algebraic degree of 2 but achieves a higher SAC score of 0.6667
- The AES S-box combines high algebraic degree with strong empirical avalanche balance. The AES S-box has algebraic degree 7, normalized degree 0.8750, and normalized SAC score 0.9473, the highest in the dataset.

6. Controlled examples and counterexamples

Case A : Sparse structural dependency

Function: Sparse Linear 1 ($4 \rightarrow 4$)

Structural metrics: Dependency density = 0.3125, normalized diffusion = 0.2500, confusion = 0.

Empirical metrics: Mean avalanche = 0.3125, mean SAC deviation = 0.5000.

Why it matters: Only 5 of the 16 possible input-output dependencies are active. Consequently, several output components remain unaffected by particular input-bit flips. Furthermore, the linear structure produces deterministic changes, resulting in maximally imbalanced SAC behaviour.

Case B : Dense dependency but poor SAC

Function: Dense Linear 1 (4 → 4)

Structural metrics: Dependency density = 0.7500, normalized diffusion = 0.5000, confusion = 1.

Empirical metrics: Mean avalanche = 0.7500, mean SAC deviation = 0.5000.

Why it matters: Despite having 12 active dependencies out of 16 and a relatively high mean avalanche probability, every input-output pair changes either always or never. This demonstrates that frequent output changes do not necessarily imply balanced avalanche behaviour.

Case C : Stronger empirical balance (if observed)

Function: Medium Quadratic 3 (4 → 4)

Structural metrics: Dependency density = 0.9375, normalized diffusion = 0.7500, confusion = 3.

Empirical metrics: Mean avalanche = 0.4688, mean SAC deviation = 0.0313.

Why it matters: Of the 16 input-output pairs, 15 exhibit perfectly balanced avalanche probabilities of 0.5, while one pair has zero dependency. This transformation demonstrates that nonlinear structure can exhibit substantially better empirical avalanche balance, although a low average SAC deviation can conceal an isolated structural deficiency.

Pairwise comparison

Although Quadratic Mix and Cubic Mix both achieve maximum dependency density (1.0000), normalized diffusion (1.0000), and confusion degree (3), their mean SAC deviations differ: 0.1667 and 0.2500, respectively.

This demonstrates that identical structural diffusion and confusion metrics do not guarantee identical empirical avalanche behaviour. Structural metrics describe potential influence, whereas SAC deviation measures how evenly that influence occurs.

7. Answering the research question

Hypothesis	Finding in this dataset	Evidence	Judgment
H1: Structural coverage relates to average avalanche	Greater dependency density generally accompanies higher mean avalanche probability, but does not uniquely determine it.	Dense Linear 1: density 0.75, mean avalanche 0.75; Mixed Cubic 3: density 0.75, mean avalanche 0.5938.	Supported with qualifications
H2: Dense coverage is insufficient for SAC	Complete structural dependency does not guarantee balanced avalanche behavior.	AES S-box and Cubic Mix both have density 1.0, but mean SAC deviations of 0.0264 and 0.2500, respectively.	Supported
H3: Confusion/ANF degree alone is insufficient	Identical confusion or algebraic degree can correspond to substantially different SAC performance.	Quadratic Mix and Cubic Mix both have confusion degree 3 but different SAC deviations. Medium Quadratic 3 and Quadratic Mix both have ANF degree 2 but different SAC scores.	Supported

The experiments indicate that structural dependency is associated with average avalanche propagation, consistent with the mathematical fact that an absent dependency cannot produce an output change. However, greater structural coverage does not necessarily imply balanced avalanche behavior. This is demonstrated by AES S-box and Cubic Mix, which achieve complete structural dependency but exhibit substantially different SAC deviations.

Similarly, structural confusion and algebraic degree do not independently determine empirical avalanche performance. Therefore, the evaluated transformations demonstrate that structural and empirical metrics provide complementary information, and neither is sufficient alone to characterize diffusion quality.

Also, Higher mean avalanche probability does not necessarily mean better SAC performance. SAC requires individual input-output flip probabilities to approach 0.5, rather than simply maximizing the frequency of output changes.

8. Limitations and Threats to Validity

- Scope: Only 28 transformations of 3, 4, and 8 bits were evaluated. Results may not generalize to larger or different function families.
- Sampling: Avalanche probabilities were computed using exhaustive input enumeration, ensuring exact results under a uniform input distribution.
- Dimension: Normalized diffusion and confusion metrics were used to compare transformations of different sizes.
- Aggregation: Average metrics may conceal poorly balanced input-output pairs; therefore, heatmaps and maximum SAC deviations were also examined.
- Metric definitions: The implemented confusion metric differs from established measures such as differential uniformity and Walsh-spectrum nonlinearity.
- Comparability: Different function families and dimensions were included without fully controlling for their differences.
- Security: Good diffusion and SAC performance alone do not guarantee cryptographic security.

9. Conclusion

This study compared structural diffusion, confusion, dependency density, and ANF degree with empirical avalanche behaviour across 28 Boolean transformations.

The results suggest that greater structural complexity can accompany improved avalanche balance, but maximum structural diffusion and confusion do not guarantee good SAC performance. For example, AES S-box and Cubic Mix both achieve complete structural dependency but have mean SAC deviations of 0.0264 and 0.2500, respectively.

The findings demonstrate that structural metrics measure the *possibility of influence*, while empirical metrics measure its *frequency and balance*. Therefore, both are necessary for a meaningful analysis of Boolean transformations.

Future work could extend the experiments to larger transformations and incorporate differential uniformity and nonlinearity measures.

References

1. Li et al. (2016). *A New Approach to the Definition of Information Diffusion and Confusion of Boolean Transformation*. Atlantis Press. <https://doi.org/10.2991/icaita-16.2016.15>
2. Golomb et al. (2002). *Claude Elwood Shannon (1916–2001)*. Notices of the American Mathematical Society. <https://www.ams.org/notices/200201/fea-shannon.pdf>